

CYBERSECURITY DASHBOARD

TECHNICAL DOCUMENTATION & USER GUIDE

Version · v2.0

Classification · UNCLASSIFIED // FOR AUTHORIZED USE ONLY

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Audience · SOC Analysts · Security Engineers · Threat Intel Teams

Format · Single-file HTML — No backend or installation required

This document provides a comprehensive reference for the Cybersecurity Dashboard, covering architecture, data sources, operational procedures, API integrations, a full glossary of CVE status terms, and recommended enhancements.

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1. Overview

The Cybersecurity Dashboard is a fully self-contained, single-file HTML application designed for Security Operations Center analysts. It aggregates real-time threat data from multiple industry-standard feeds into a unified, actionable interface requiring no backend infrastructure, no server, and no installation.

Key Characteristics

- **Zero-dependency** — runs entirely in the browser from a single .html file
- **Live data** — pulls from 5+ public and authenticated threat intelligence APIs
- **No backend** — API keys stored locally in the browser; never sent to any proxy
- **Responsive** — Desktop analyst view and large-font TV display mode
- **Analyst-friendly** — built-in research modal, patch queue, and one-click IOC copy
- **Auto-refresh** — configurable polling interval (default: 15 minutes)

2. Getting Started

Prerequisites

Requirement	Details
Web Browser	Chrome 90+, Firefox 88+, Edge 90+, or Safari 15+
Internet Access	Required — all data is fetched live from external APIs
OTX API Key	Free registration at otx.alienvault.com — optional but recommended
Screen Resolution	1280px or wider recommended; 1920px+ ideal for TV mode

Initial Setup

Step 1	Open the file — Double-click <code>cyber_threat_dashboard.html</code> or drag it into your browser.
Step 2	Enter OTX Key — Paste your AlienVault OTX API key. Unlocks threat pulses, actors, and IOC feeds.
Step 3	Choose interval — Select auto-refresh frequency: 5 / 15 / 30 / 60 min or manual only.
Step 4	Display mode — Desktop for workstations, TV for large wall displays.
Step 5	Launch — Click INITIALIZE CYBERSECURITY DASHBOARD . All feeds begin loading.

Note: NVD, CISA KEV, EPSS, and News feeds work without any API key. The OTX key is only needed for threat pulses, actor profiles, and IOC feeds.

3. Dashboard Interface

Header & Controls

Element	Description
Logo / Title	CYBERSECURITY DASHBOARD — primary branding identifier
LIVE badge	Pulsing green indicator confirming active data fetching
Refresh button	Manually trigger an immediate refresh of all feeds
Interval selector	Change auto-refresh interval at any time without reloading
Desktop / TV toggle	Switch between compact analyst and large-font TV display modes
Last fetch	Timestamp of the most recently completed data refresh

Timezone Bar

A dedicated bar beneath the header displays two live clocks updating every second — LOCAL time (the browser's local timezone) and UTC. Both clocks use a fixed-width Orbitron font so the panel does not resize as seconds change.

Metric Cards

Card	Color	Metric	Source
CRITICAL CVEs	Red	Count of CVSS 9.0+ vulnerabilities	NIST NVD
HIGH CVEs	Amber	Count of CVSS 7.0–8.9 vulnerabilities	NIST NVD
INCIDENTS	Blue	Active CISA KEV entries	CISA KEV
ACTORS	Green	Adversary groups tracked	AlienVault OTX
KEV ENTRIES	Purple	Total known-exploited vulnerabilities	CISA KEV
THREATS	Cyan	Active OTX community threat pulses	AlienVault OTX

CVE Severity Distribution Chart

A doughnut chart displays the breakdown of the current NVD fetch by severity. Four severity slices (Critical, High, Medium, Low) are shown with color-coded segments. Hovering over a slice shows the count and percentage in a tooltip. Compact stat tiles to the right of the chart show each severity's percentage at a glance. The chart updates on every data refresh.

4. Data Feeds & Tabs

Tab	Source	Description	Key?
CVEs	nvd.nist.gov	Recent vulnerabilities from the National Vulnerability Database	No
KEV	cisa.gov	CISA Known Exploited Vulnerabilities with search filter	No
OTX	otx.alienvault.com	Community threat pulses and adversary actor profiles	Yes
IOC	otx.alienvault.com	Malicious IPs and domains from OTX pulses	Yes
EPSS	api.first.org	Exploit Prediction Scoring — likelihood of exploitation in 30 days	No
Patch	NVD+EPSS+KEV	Prioritized patch queue; click rows to mark done	No
News	RSS	Aggregated news from BleepingComputer, THN, Krebs, SecurityWeek	No
Map	OTX	Geographic map of threat actor origin countries	Yes

CVE List (NVD Tab)

All scored CVEs are displayed sorted by CVSS score (highest first). Unscored CVEs — those labeled N/A, meaning NIST has not yet assigned a CVSS score — appear at the bottom of the list. This is normal for recently published CVEs that are still awaiting analysis. Click any row to open the Research Modal.

Patch Priority Queue

Priority	Criteria	Target SLA
CRITICAL	CVSS >= 9.0 AND present in CISA KEV	24 hours
HIGH	CVSS >= 7.0 OR EPSS score >= 50%	72 hours
MEDIUM	All other CVEs in the NVD fetch window	30 days

5. API Reference

API / Service	Endpoint	Auth	Rate Limit
NIST NVD	services.nvd.nist.gov/rest/json/cves/2.0	None	5 req/30s
CISA KEV	cisa.gov/.../known_exploited_vulnerabilities.json	None	No limit
FIRST EPSS	api.first.org/data/v1/epss	None	No limit
OTX Pulses	otx.alienvault.com/api/v1/pulses/subscribed	X-OTX-API-KEY	10k/day
OTX Actors	otx.alienvault.com/api/v1/pulses/related	X-OTX-API-KEY	10k/day
OTX IOC	otx.alienvault.com/api/v1/indicators/export	X-OTX-API-KEY	10k/day
RSS / News	Various via corsproxy.io	None	Proxy-dependent

Note: A CORS proxy is required because browsers block direct cross-origin API requests. The dashboard uses corsproxy.io with allorigins.win as a fallback.

6. Severity Classification

Severity	CVSS v3.1 Range	Color	Recommended SLA
CRITICAL	9.0 – 10.0	Red	Immediate — within 24 hours
HIGH	7.0 – 8.9	Amber	Urgent — within 72 hours
MEDIUM	4.0 – 6.9	Blue	Standard — within 30 days
LOW	0.1 – 3.9	Green	Scheduled — next patch cycle
N/A	Not yet scored	Grey	Monitor — pending NVD analysis

N/A items are CVEs that have been published by NVD but have not yet received a CVSS score. These are sorted to the bottom of the CVE list and should be monitored as scores are typically assigned within days to weeks of publication.

7. CVE Status Glossary

When the NVD publishes a CVE, it moves through several workflow states before full analysis is complete. The dashboard displays these statuses in the CVE list so analysts understand how much information is available and how reliable the data is.

Awaiting Analysis

Definition: The CVE record has been received and reserved by NVD but has not yet been reviewed by a NVD analyst. No CVSS score, CPE (affected product), or CWE (weakness type) data has been assigned yet.

Analyst Context: CVEs in this state will show as N/A in the dashboard severity column. This is the most common status for recently published CVEs — NVD typically processes hundreds of new entries per week and a backlog is normal. Do not assume low risk simply because no score has been assigned; check vendor advisories directly for preliminary severity guidance.

Recommended Actions:

- Check the vendor's own security advisory for an interim CVSS score
- Search for the CVE ID in OTX pulses — community intelligence may already be available
- Monitor EPSS — a rising exploitation probability score is an early warning even before NVD scores the CVE

Undergoing Analysis

Definition: NVD has begun the formal review process for this CVE. An analyst is actively evaluating the vulnerability to assign CVSS metrics, affected CPE configurations, and CWE classification. A score may be published within hours or days.

Analyst Context: This status is transitional and brief. CVEs here may still appear as N/A in the dashboard if the score has not yet been published. Treat these with the same caution as "Awaiting Analysis" until the score is confirmed.

Recommended Actions:

- Re-check the dashboard on the next refresh cycle — scores are often published within 24-48 hours
- Cross-reference with CISA KEV; if the CVE appears there, it is being actively exploited regardless of NVD status

Analyzed

Definition: NVD analysis is complete. The CVE has been assigned a CVSS base score, affected product configurations (CPE), and a weakness category (CWE). This is the fully enriched state and represents the most reliable data.

***Analyst Context:** Analyzed CVEs will display their full CVSS score and severity badge in the dashboard. All patch prioritization, EPSS cross-referencing, and KEV lookups function correctly for CVEs in this state. The majority of CVEs in the NVD feed at any given time will be in the Analyzed state.*

Recommended Actions:

- Use the CVSS score combined with the EPSS score to prioritize remediation
- Check if the CVE appears in CISA KEV — if so, exploitation is confirmed in the wild
- Click the CVE row to open the Research Modal and access NVD, Shodan, and GitHub PoC links

Modified

Definition: A previously Analyzed CVE has been updated. This can occur when new information emerges — such as additional affected products, a revised CVSS score, or correction of an earlier analysis error. The modification date will differ from the publication date.

***Analyst Context:** A Modified status on a CVE that was previously scored LOW or MEDIUM is a significant alert — re-scoring to HIGH or CRITICAL after initial publication is not uncommon as exploitation details emerge. The dashboard sorts by score so re-scored CVEs will automatically rise to the top of the list on the next refresh.*

Recommended Actions:

- Pay particular attention to Modified CVEs whose severity increased since last refresh
- Re-evaluate patch priority if a CVE in your environment has been re-scored upward

Deferred

Definition: NVD has determined that this CVE will not be fully analyzed in the near term, typically because the vulnerability is in a product not in scope for NVD's standard enrichment process, or because additional information is needed from the vendor or reporter before analysis can proceed.

***Analyst Context:** Deferred CVEs may remain unscored indefinitely. This does not mean the vulnerability is not serious — it means NVD cannot complete analysis with currently available information. Deferred CVEs appear as N/A in the dashboard and should be investigated directly through vendor channels. CISA KEV entries can be Deferred in NVD yet still represent actively exploited vulnerabilities.*

Recommended Actions:

- Do not treat Deferred as low risk — check vendor advisories and CERT bulletins
- Search for the CVE in OTX and threat intelligence platforms for community-sourced risk data
- If the vulnerability affects software in your environment, apply vendor patches regardless of NVD status
- Check CISA KEV directly — a CVE can be Deferred in NVD and still appear in KEV as actively exploited

Rejected

Definition: The CVE ID has been officially rejected and should not be used. This happens when a CVE was reserved but the reported vulnerability was found to be a duplicate of an existing CVE, was not a genuine vulnerability, or was reported in error.

Analyst Context: Rejected CVEs are rarely returned by the NVD API in standard queries but may appear in threat intelligence feeds. If you encounter a Rejected CVE in any feed, it should be dismissed and the correct CVE ID substituted if one exists.

Recommended Actions:

- If you see a Rejected CVE in IOC or threat pulse data, identify the canonical CVE ID using NVD search
- Remove Rejected CVEs from any watchlists, SIEM rules, or patch tickets

8. Research Modal

Clicking any item in the CVE, KEV, OTX, News, or Actor tabs opens a research modal. The modal provides curated external research links tailored to the item type.

Item Type	Research Links
CVE / Vulnerability	NVD Entry · CVEdetails · GitHub PoC search · Google (exploit/patch) · Shodan · MITRE ATT&CK;
Threat Actor	MITRE ATT&CK; · Google · Malpedia · OTX search · Mandiant · Recorded Future
News / General	Google · BleepingComputer · The Hacker News · Krebs on Security · SecurityWeek

9. Status Bar & Alerting

Element	Description
Connection dot	Green pulsing = feeds active. Red = error. Amber = partial failure.
Status label	READY / FETCHING / ERROR — current operational state
Log line	Latest event: fetch completed, error detail, or refresh trigger
Next refresh	Live countdown to the next automatic refresh cycle
Last fetch	Timestamp of the most recently completed full data refresh

CRITICAL CVE alerts (CVSS $\geq$ 9.0) trigger a slide-in notification in the top-right corner. Alerts do not auto-dismiss, remaining visible on unattended TV displays until manually closed.

10. Display Modes

Property	Desktop Mode	TV Mode
Base font	14px	20px
Metric numbers	28px Orbitron	44px Orbitron
Chart height	130px	190px
Best for	Analyst workstations, laptops	1080p / 4K wall displays
Persistence	Saved in localStorage (soc_mode)	Saved in localStorage (soc_mode)

11. Recommended Enhancements

High Priority

Enhancement	Description
Triage Tags	Per-item status labels (Investigating / Escalated / False Positive / Resolved) persisted in localStorage.
Export CSV / JSON	Download the patch queue, IOC list, or CVE feed for SIEM ingestion or reporting.
Offline Cache	Store the last successful API response in localStorage so the dashboard works when APIs are rate-limited.
CORS Proxy Fallback	Fallback chain (corsproxy.io → allorigins.win) to prevent blank panels when the proxy is down.

Medium Priority

Enhancement	Description
Analyst Notes	Freetext annotation per CVE or actor, stored in localStorage, visible in lists and research modal.
Push Notifications	Browser Notifications API to surface CRITICAL CVE alerts even when the tab is in the background.
EPSS vs CVSS Chart	Scatter plot of CVSS (X) vs EPSS (Y). High-EPSS / high-CVSS quadrant is the immediate action zone.
KEV Cross-reference	Badge CVEs in the NVD list that also appear in CISA KEV so confirmed exploited status is visible without switching tabs.

12. Troubleshooting

Problem	Panels show "fetching..." and never load
Solution	Check internet connection. Corporate proxies may block API calls. corsproxy.io may be temporarily down — wait and click Refresh.
Problem	OTX feeds show no data
Solution	Verify your OTX API key at otx.alienvault.com. Use the Forget button to re-enter. OTX enforces 10,000 requests/day.
Problem	News feed is empty
Solution	RSS feeds route through corsproxy.io. If the proxy is down, all four news sources fail. Click Refresh on the News tab header.

Problem	EPSS shows 0 results
Solution	EPSS cross-references CVE IDs from the NVD fetch. If NVD failed, EPSS has no IDs. Refresh CVEs tab first, then retry EPSS.
Problem	Patch checkboxes reset
Solution	Checked state is stored in localStorage (soc_patched). Clearing browser data, incognito mode, or a different browser profile will reset all checkboxes.
Problem	Low CVEs not visible in the list
Solution	Low and N/A CVEs sort below scored items. If more than 40 higher-severity CVEs are loaded, scroll down to find them. The dashboard now shows all scored CVEs plus up to 20 unscored items.

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